v2 models for data assimilation

Shared encoder. Distinct decode.

A_CRPS

PatchConvMLP

$9 \rightarrow 32$

native-z PCA

HeaveFast

HeaveResidualFast

$11 \rightarrow 35$

warp + residual PCA

ops

HeaveResidualFast

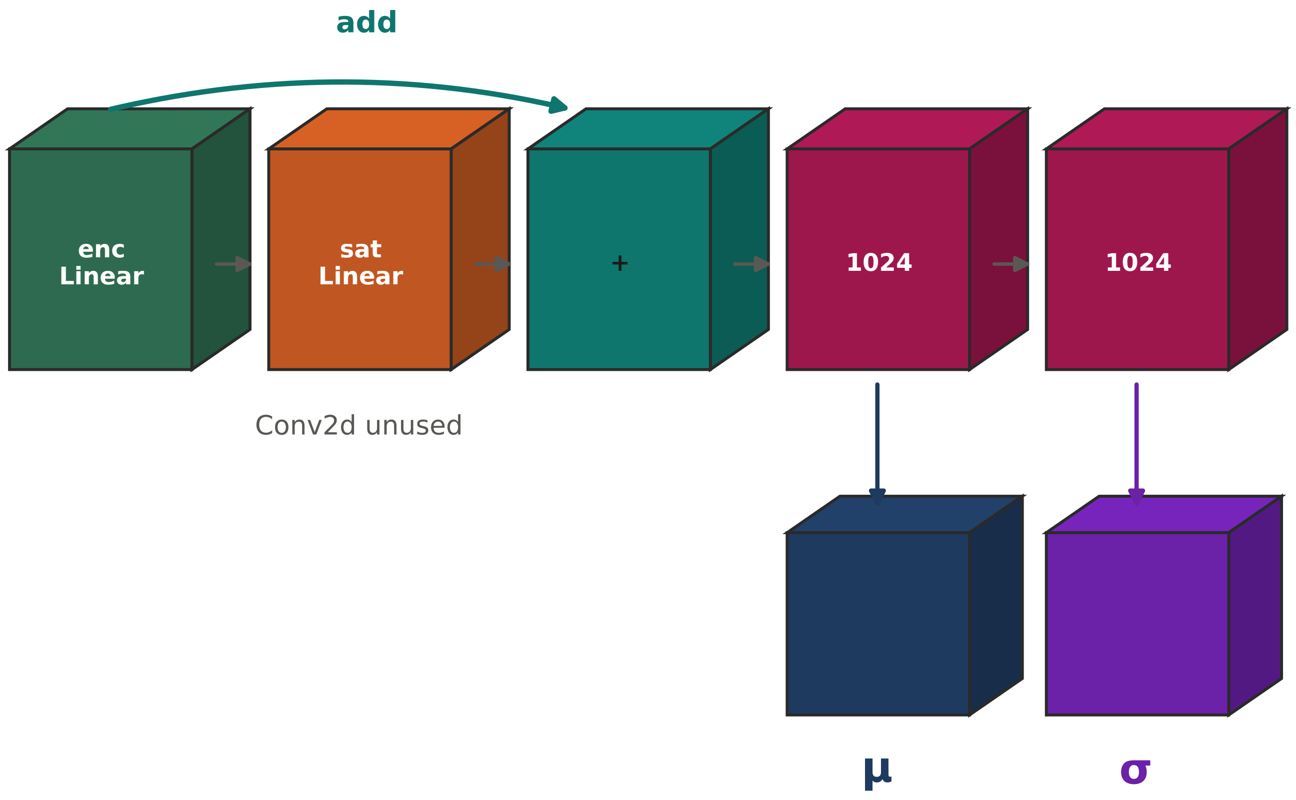
$30 \rightarrow 35$

warp + 19 operators

PatchConvMLP

patch_shape is None. Satellite encoding is Linear. Conv2d is unused.

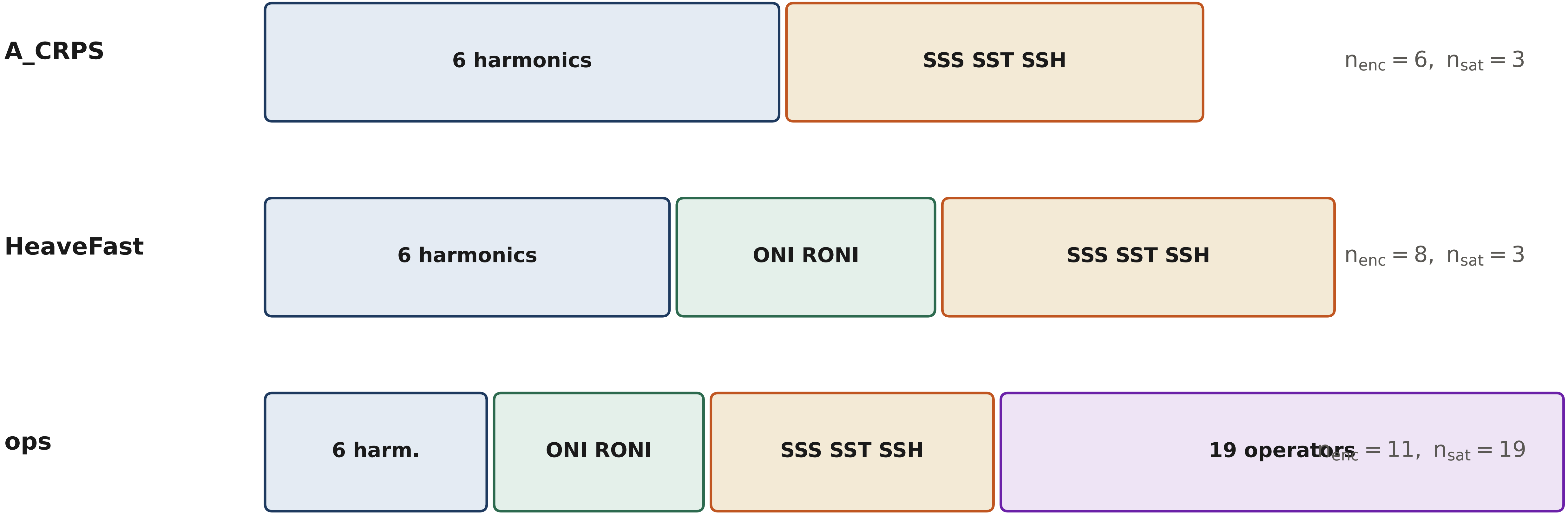
```
enc = x[:, :n_enc]
h_e = Linear(n_enc, 128)(enc)
sat = x[:, n_enc:]
h_s = Linear(n_sat, 128)(sat)
h = h_e + h_s              # conv is None
h = Drop(ReLU(Linear(128,1024)(h)))
h = Drop(ReLU(Linear(1024,1024)(h)))
μ = Linear(1024, d)(h)
σ = softplus(Linear(1024,d)(h))+ε
```



$$h_0 = W_e x_{enc} + W_s x_{sat}$$

Input layout

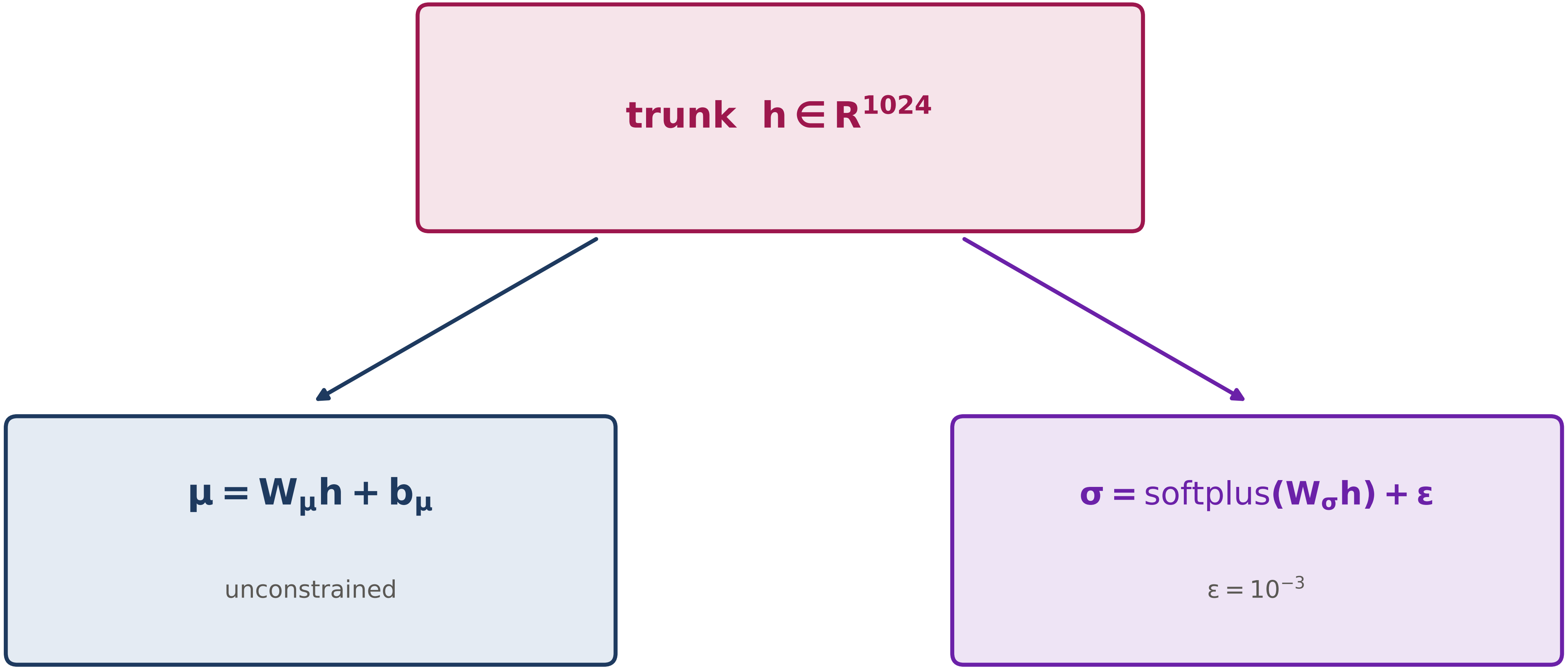
Harmonics, optional ENSO, then satellite columns.



$$(\cos,\sin)\,2\pi\,\mathrm{doy}/365,\quad (\cos,\sin)\,2\pi\,\mathrm{lat}/180,\quad (\cos,\sin)\,2\pi\,\mathrm{lon}/360$$

Heteroscedastic head

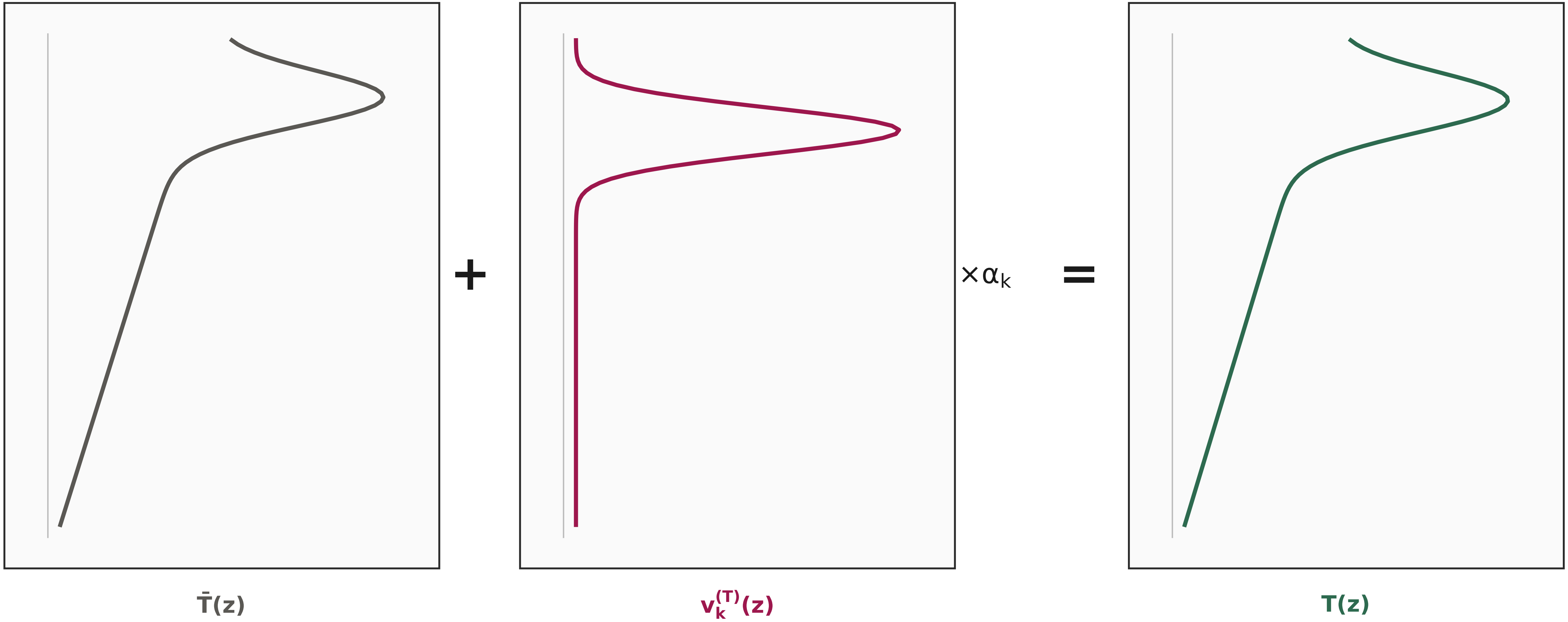
Only μ is assimilated.



A_CRPS reconstruction

Native 1 m grid. 16 T and 16 S principal components.

$$T(z) = \bar{T}(z) + \sum_{k=1}^{16} \alpha_k v_k^{(T)}(z)$$

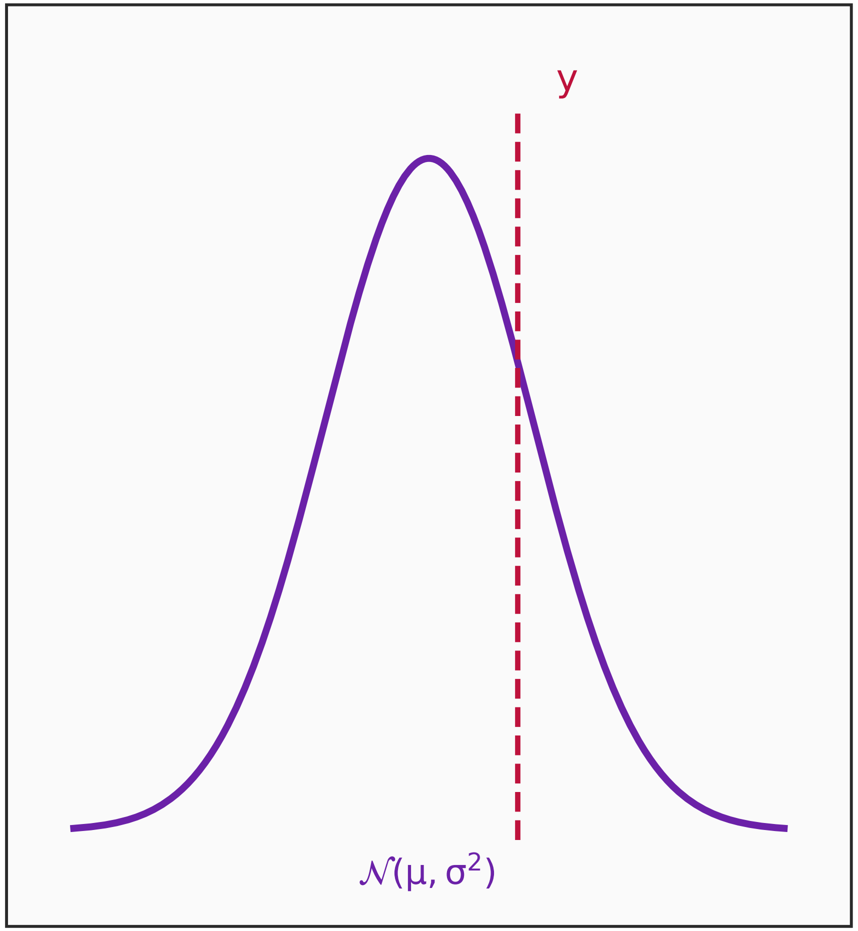


CRPS

Independent Gaussian scores. No predicted covariance.

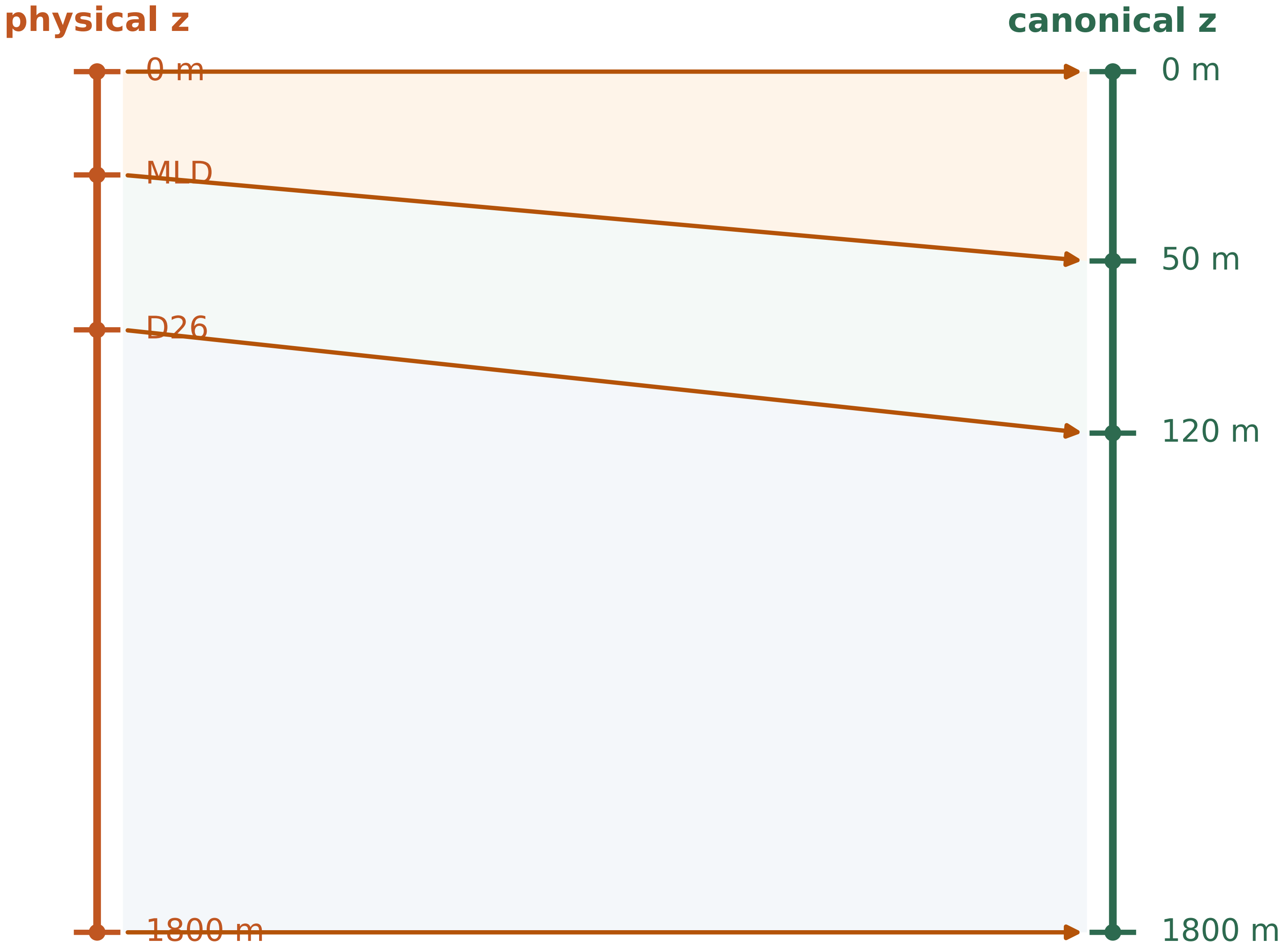
$$\mathrm{CRPS} = \sigma[z(2\Phi(z) - 1) + 2\phi(z) - \pi^{-1/2}]$$

$$z = (y - \mu)/\sigma$$



Heave warp

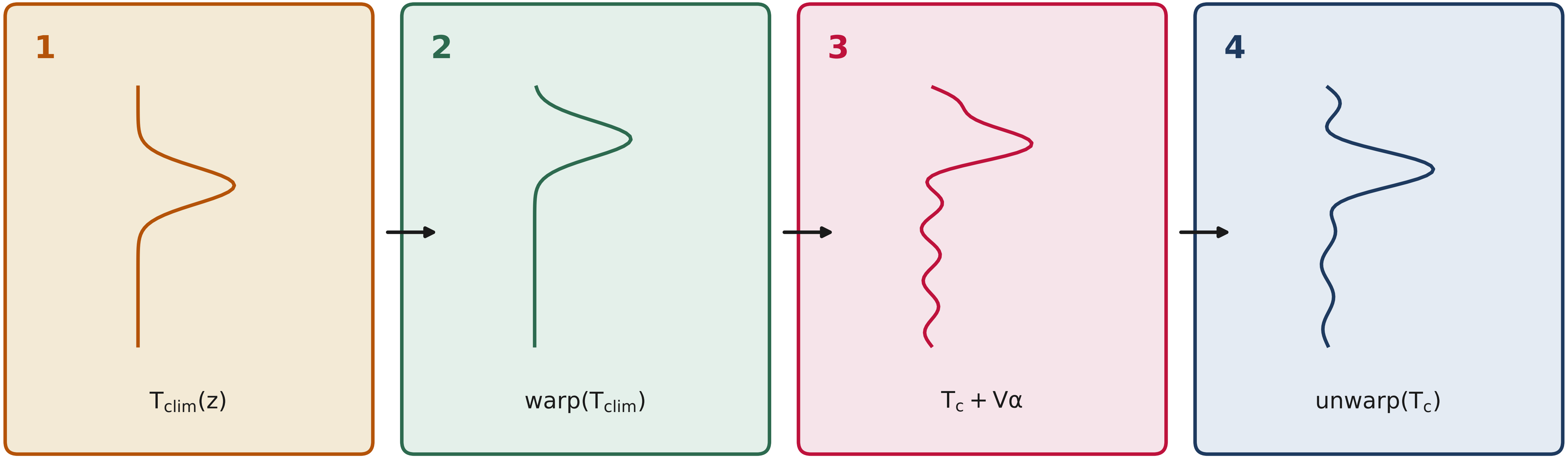
Piecewise-linear map of MLD and D26 onto a canonical grid.



$$\mathrm{MLD} = 50\,e^{r_0}, \quad \mathrm{D26} = \mathrm{MLD} + 5 + 65\,e^{r_1}$$

Heave residual

Warp climatology, add residual PCA, invert the map.



$T_{res} = \alpha_T V_T + \bar{m}_T$ (train-split warped residuals)

Heave loss

Geometry, CRPS, and a weak 50–200 m Tz term.

$$L = 10\,L_{\mathrm{geom}} + L_{\mathrm{CRPS}} + 0.1\,L_{\partial_z T}$$

L_{geom}

MLD/50, D26/120

L_{CRPS}

MLD, D26, 32 PCs

$L_{\partial_z T}$

50–200 m

ops features

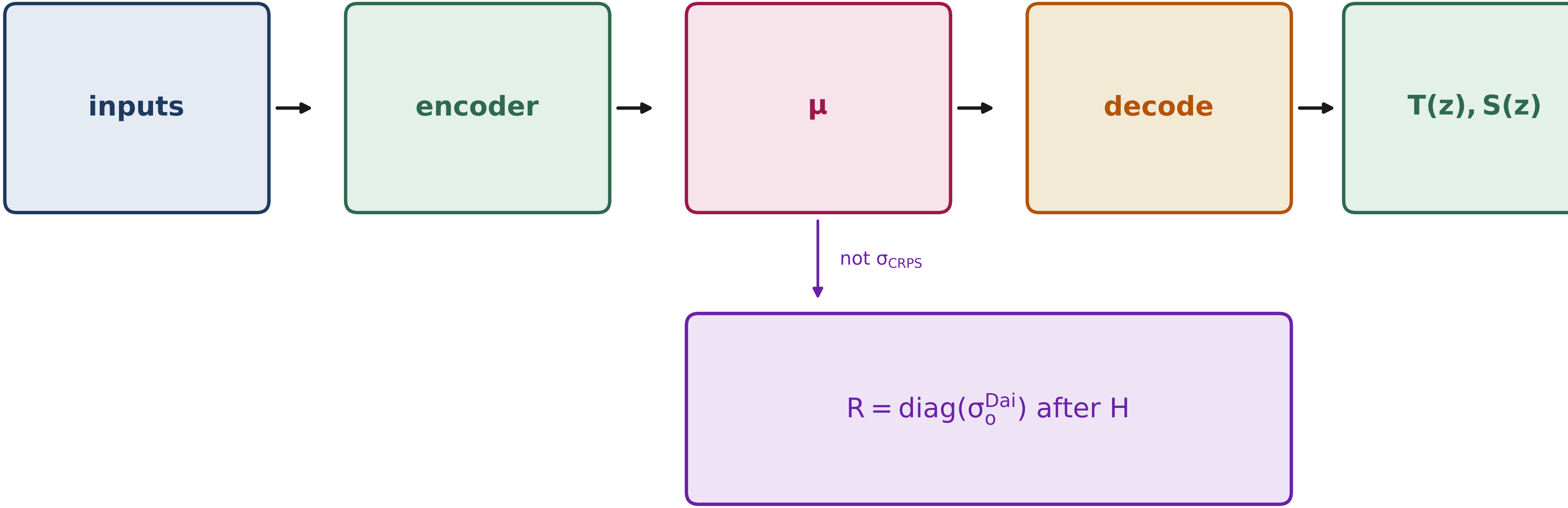
Nineteen operators sampled from the regional cube.

sst.gx loc	sst.gy loc	sst.gx 1°	sst.gy 1°	sss.gx loc	sss.gy loc	sss.gx 1°
sss.gy 1°	ssh.gx loc	ssh.gy loc	ssh.gx 1°	ssh.gy 1°	$\nabla^2\eta$	sst 7d
ssh 7d	u_g loc	v_g loc	u_g 1°	v_g 1°		

$$u_g = -(g/f)\partial_y\eta, \quad v_g = (g/f)\partial_x\eta$$

Assimilation products

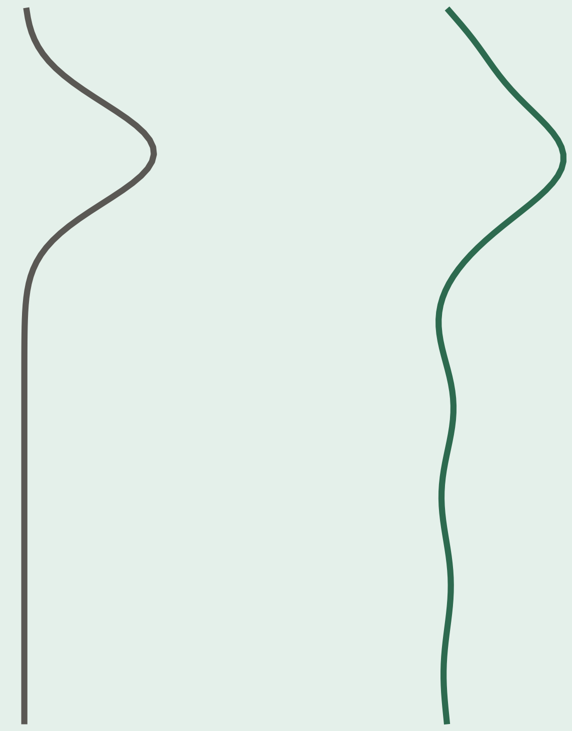
Profiles from μ. Diagonal R from Dai σ_o after H.



Two reconstructors

A_CRPS is additive in z. Heave is additive after a vertical map.

A_CRPS



Heave